- Let’s start with my introduction. As you are aware my name is …… I have .. years of experience. I started my career as a trainee in which My tasks were to monitor the infrastructure and applications. There I was working with PRTG as a Monitoring tool.

- Talking about cloud providers I have an experience working with all 3 leading cloud providers. GCP, Azure and AWS.

- In AWS services I have knowledge of EC2, IAM, VPC, S3, Session Manager, Route53, Cloud trail, Cloud Watch, Cloud front, SNS, SQS, Load Balancer (LB), Lambda, MSK, EKS.

- In Azure I have knowledge of Azure Virtual Machines (VMs), Azure Kubernetes Service (AKS), Azure Disk Storage, Azure File Storage, Azure Content Delivery Network (CDN)

- In GCP I have knowledge of Google Compute Engine (GCE), Google Kubernetes Engine (GKE), Google Cloud Storage (GCS), Cloud Monitoring, Cloud Logging, Cloud Trace.

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- Apart from that I have knowledge of Linux, DataDog as a monitoring Tool, GIT as version control system, Github for repository control & GitLab for Repository and CI/CD pipeline.

- I closely work with EKS(AWS) and AKS(Azure) for upgradation of the Kubernetes version and manage the infrastructure with the terraform.

- Apart from the in my day today activities I have to manage the confluent cloud that is the service of Kafka and MSK that is Kafka service from AWS. Setup the monitoring system of the EKS and Confluent in Datadog. I have to manage the monitors and Dashboards so that if there is any issue with any application it can easily be traced and troubleshooted as soon as possible, so that we can meet our SLO(Service level objective).

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Real Time issues in Linux

- Log rotate fail
- Cronjob
- Disk full
- Inode disk full
- /etc/fstab issues (Related to mounting)

Real Time work in Linux

- Upgrading the centos version
- kernel upgrade
- Mount disks
- Patching by finding the vulnerabilities using Rapid7

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Real Time issues in Kubernetes

- **Pods gets failed due to resource requests and limits**

Fix - **isolated CPU cores** to ensure that real-time applications run on dedicated CPU cores, avoiding interference from non-real-time workloads.

- **Pods running on different nodes.**

  Fix - Set **priority classes** to ensure that critical pods are scheduled preferentially.

- **Out-of-memory (OOM) kills**

  Symptoms - Pods might get killed if the node runs out of memory, leading to sudden termination of real-time applications.

  Fix - Ensure **resource limits** and **requests** are set correctly to avoid OOM kills.

Use **Pod Disruption Budgets (PDB)** to control the minimum availability of critical pods during voluntary disruptions.

- **Network issues**

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Realtime Issues with Terraform

- **State file conflicts:** If multiple users or automation systems are working on the same infrastructure, state file conflicts can occur.
- **Remote state latency**: If using a remote state backend (e.g., S3, Azure Blob Storage), network latency or issues with the remote state backend can delay infrastructure updates.
- **Manual interventions**: Infrastructure modified outside of Terraform (e.g., manual changes in the console, CLI, or other automation systems) creates state drift.
- **Version constraints**: Sometimes there is version constraint in State file. Mean some state files were made using 1.5.7 but we are trying to run with 0.13.7 I USE TFSWITCH to change the tf version easily.

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Real time Issues in Kafka

1. Latency

Network delays.

- Inefficient producers (e.g., slow message publishing).
- Network delays.
- Consumer lag (e.g., when consumers are not fast enough).

Solution:

- Optimize producer configurations (e.g., batch size, compression).
- Tune the consumer side (e.g., increase consumer parallelism, use efficient deserialization methods).
- Use Kafka's "acks" settings appropriately to balance durability and latency

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2. Consumer Lag

- **Issue:** Consumers falling behind the producers, creating lag.
- **Causes:**
  - Insufficient consumer resources.
  - Slow processing of messages.
  - Overloaded Kafka brokers.
- **Solution:**
  - Monitor consumer lag with Kafka's consumer lag metrics.
  - Scale consumer groups horizontally by adding more consumers to the group.
  - Optimize the consumers' processing logic and the Kafka broker setup.

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3. Data Ordering and Message Duplication

- **Issue:** Kafka provides "at least once" or "exactly once" message delivery semantics, but in some configurations, message duplication or out-of-order messages can occur.
- **Causes:**
  - Producer retries.
  - Network issues or broker crashes during message processing.
  - Faulty producer logic.
- **Solution:**
  - Enable "idempotent" producer settings to avoid duplicate messages.
  - Configure message delivery guarantees correctly ("acks" settings).
  - Use Kafka Streams or other frameworks that ensure message ordering in certain use cases.

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4. Disk I/O Bottlenecks

- **Issue:** Kafka relies on disk I/O for storing and retrieving data. If the disks become a bottleneck, performance suffers.
- **Causes:**
  - Slow disk drives (e.g., HDDs instead of SSDs).
  - High traffic and data volume.
  - Poor disk management.
- **Solution:**
  - Use high-performance SSDs for Kafka brokers.
  - Adjust disk configurations (e.g., configure disk partitions with high throughput).
  - Implement proper log segment management.

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5. Partition Imbalance

- **Issue:** Kafka's partitioning mechanism might cause uneven distribution of data across brokers, leading to some brokers being overwhelmed while others are underutilized.
- **Causes:**
  - Poor partitioning strategy (e.g., using an inefficient key for partitioning).
  - Insufficient number of partitions.
- **Solution:**
  - Use a well-defined partition key that distributes data evenly.
  - Rebalance partitions using Kafka's partition reassignment tool.
  - Ensure the number of partitions is adequate for your throughput and parallelism needs.

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6. Network and Connectivity Issues

- **Issue:** Kafka is sensitive to network connectivity problems, especially when handling real-time data streams.
- **Causes:**
  - Network congestion or packet loss.
  - Poorly configured network infrastructure.
- **Solution:**
  - Ensure low-latency and high-throughput networking.
  - Use Kafka's internal monitoring to detect and resolve network issues early.

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7. Throughput Bottlenecks

- **Issue:** Kafka may not be able to handle high throughput workloads efficiently.
- **Causes:**
  - Insufficient broker resources (CPU, memory, disk space).
  - Inefficient producer and consumer configurations.
  - Poor network or I/O performance.
- **Solution:**
  - Scale Kafka clusters horizontally.
  - Tune producer configurations to use batching and compression.
  - Ensure that network infrastructure can handle the expected traffic.

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Best Practices for Real-Time Kafka Systems:

- **Monitoring & Alerting:** Continuously monitor system metrics (consumer lag, disk usage, network throughput, etc.) and set up alerts for any critical thresholds.
- **Partitioning Strategy:** Design an effective partitioning strategy to ensure even load distribution across brokers and allow parallel consumption.
- **Replication & Fault Tolerance:** Ensure proper replication for fault tolerance and minimize the impact of broker failures.
- **Scalability:** Plan for horizontal scaling both at the broker level and at the consumer level to handle growth in traffic.

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